

IT443 Resampling Techniques and Bayesian Computation (Winter '2026)

Instructor: Subhasish Basak, 3206, Faculty Block 3

Course Placement: B.Tech. (ICT), Semester-VI

Course Format: 3 Lecture hours + 2 Lab hours (3-0-2)

Course Content: This course provides a rigorous introduction to resampling techniques and Bayesian computation. The resampling module covers jackknife, bootstrap, and variance estimation. The Bayesian module covers priors, posteriors, conjugacy, credible intervals, and MCMC methods including Metropolis–Hastings and Gibbs sampling.

Textbooks:

1. **An Introduction to the Bootstrap** — Efron & Tibshirani
2. **Computer Age Statistical Inference** — Efron & Hastie
3. **Bayesian Data Analysis** — Gelman et al.
4. **Monte Carlo Statistical Methods** — Robert & Casella

Evaluation Plan:

1. Two Assignments – 20%
2. In-Semester Exam I – 20%
3. In-Semester Exam II – 20%
4. End-Semester Examination – 40%

Course Outcomes:

CO1: Apply fundamental probabilistic concepts, resampling methods, and Bayesian inference techniques to model, analyze, and draw conclusions from real-world data.

CO2: Use modern computational tools to design and evaluate solutions for real-world systems.

CO3: Communicate findings effectively, work in teams, and undertake lifelong learning by exploring advanced topics.

POs-COs Matrix: (✓✓ strong ✓ moderate • introductory – none)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓	✓	–	•	–	–	–	–	–	–	–	•
CO2	✓✓	•	–	•	✓✓	–	–	–	–	–	•	•
CO3	•	•	–	–	–	–	–	–	✓	✓	✓	✓✓

Tentative Lecture Distribution:

Topic Name	Content (2 -3 lines per 4 – 6 lectures)	No. of lectures
Basic Probability	Quick refresh of probability basics. Joint and conditional probabilities, common distributions, expectation and variance. Conditional expectation, law of total expectation, simple examples to warm up.	4
Resampling techniques	Brief introduction to resampling, Examples of calculating variance, sampling distributions. Different resampling methods: Jackknife, Bootstrap, Half sampling etc.	4
Bootstrap and Jackknife for i.i.d cases	Bootstrap in parametric and non-parametric setups. Consistency properties, comparison of bootstrap with normal approximation. Jackknife for iid case, consistency, comparison with non-parametric boot strap	8
Bootstrapping on linear models	Residual Bootstrap, weighted bootstrap, discussion about robust and efficient bootstrap techniques.	4
Brief introduction to Bayesian Inference	Prior and posterior distributions, Conjugacy, Non-informative priors, Posterior mean, variance, credible intervals and high-posterior density intervals, comparison between Bayesian and frequentist techniques	8
Bayesian Bootstrap	Comparison of Bayesian credible interval with bootstrap confidence intervals. Bayesian Bootstrap, Introduction to	8

and MCMC	MCMC, Metropolis-Hastings and Gibbs sampling,	
Applications	Case studies linking theory with practice. Examples from physics, biology, and statistics. Discussions and problem-solving sessions.	4
Total		40